0580_m24_ms_42_Q5

Question	Answers	Marks	Partial Marks
5(a)(i)	$(x-4)(x+3)$ final answer	**2**	**M1** for $(x+a)(x+b)$ where $ab=-12$ or $a+b=-1$ or for $x(x+3)-4(x+3)$ or $x(x-4)+3(x-4)$
5(a)(ii)	$\frac{x+4}{x+3}$ final answer	**2**	**M1** for $(x-4)(x+4)$ seen
5(b)	$3x^2-14x+8$ or $(x-4)(3x-2)$ final answer	**3**	**M2** for $((2x-3)-(x+1))((2x-3)+(x+1))$ or $(4x^2-6x-6x+9)-(x^2+x+x+1)$ or better or correct answer seen or **M1** for $(x-4)(ax+b)$ or $(3x-2)(x+c)$ or $(4x^2-6x-6x+9)$ or $\pm(x^2+x+x+1)$ oe

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Question	Answers	Marks	Partial Marks
5(c)	$\frac{x^2-3x-12}{(x+1)(x-3)}$ or $\frac{x^2-3x-12}{x^2-2x-3}$ final answer	**4**	**B1** for common denominator $(x+1)(x-3)$ oe isw **B1** for $(2x+4)(x-3)-x(x+1)$ or better seen **B1** for $2x^2-6x+4x-12$ or $-x^2-x$ seen
5(d)	$2x^3-15x^2+22x+15$ final answer	**3**	**B2** for correct expansion of three brackets unsimplified or for simplified four-term expression of correct form with 3 terms correct in final answer or **B1** for correct expansion of two brackets with at least 3 terms out of 4 correct

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Question	Answers	Marks	Partial Marks
5(e)	$2x^2 - 3x - 77[=0]$ oe $(6x^2 - 9x - 231[=0])$ or $18y^2 + 147y + 222[=0]$ oe $(6y^2 + 49y + 74[=0])$	**M2**	**M1** for correct method to eliminate one variable e.g. $2(13+3y)^2 - 9y = 116$ or $2x^2 - 3(x-13) = 116$ oe
	$(2x+11)(x-7)\ [=0]$ oe or $\dfrac{[--]3 \pm \sqrt{([-]3)^2 - 4\times2\times-77}}{2\times2}$ oe or $(6y+37)(3y+6)\ [=0]$ or $\dfrac{-147 \pm \sqrt{147^2 - 4\times18\times222}}{2\times18}$ oe	**M2**	**FT** *their* 3-term quadratic in x or y, correct factors, correct substitution into formula or for correctly completing square **M1** for a pair of factors giving 2 correct terms when expanded *their* quadratic or for e.g. $\sqrt{([-]3)^2 - 4\times2\times-77}$ oe or $\dfrac{[--]3 \pm \sqrt{p}}{2\times2}$ oe
	$x = 7$ and $y = -2$ $x = -5\frac{1}{2}$ oe and $y = -6\frac{1}{6}$ oe	**B2**	**B1** for both x-values or both y-values or for 1 correct pair